

Unit-I

INTRODUCTION TO BIG DATA ANALYTICS

Overview of Big Data Analytics:

Big Data Analytics refers to the process of examining large and varied data sets—commonly known as "big data"—to uncover hidden patterns, correlations, market trends, and other valuable insights that can help organizations make informed business decisions. This process involves collecting, organizing, and analysing massive data sets that are too complex for traditional data-processing software.

1. Types of Big Data Analytics

Big Data Analytics can be categorized into different types based on the goals of the analysis:

- **Descriptive Analytics:** Focuses on summarizing historical data to understand what has happened in the past. Tools such as dashboards and reports are commonly used here.
- **Diagnostic Analytics:** This analysis explains why something happened by identifying the causes of trends or anomalies in historical data.
- **Predictive Analytics:** Uses statistical models and machine learning techniques to predict future outcomes based on historical data.
- **Prescriptive Analytics:** Recommends actions or strategies that could be taken to achieve desired outcomes based on predictive data models.

2. Key Technologies in Big Data Analytics

- **Data Storage:** Managing vast amounts of data requires advanced storage solutions such as Hadoop Distributed File System (HDFS) and cloud storage platforms.
- **Data Processing:** Tools like Apache Hadoop, Apache Spark, and NoSQL databases (e.g., Cassandra, MongoDB) are popular for handling big data. These tools allow for parallel processing of data across multiple nodes.
- **Data Visualization:** Visualization tools like Tableau, Power BI, and D3.js help present complex insights in an understandable and actionable format for decision-makers.
- **Machine Learning:** Predictive analytics often relies on machine learning algorithms to model data. Popular frameworks include TensorFlow, Scikit-learn, and PyTorch.

3. Applications of Big Data Analytics

- **Big Data Analytics has transformative effects across multiple industries:**
- **Healthcare:** Analysis of medical records, treatment data, and research papers for disease detection, drug discovery, and personalized medicine.
- **Finance:** Fraud detection, risk management, and real-time stock market predictions.

- Retail: Customer behavior analysis, personalized marketing, inventory management, and demand forecasting.
- Manufacturing: Predictive maintenance, supply chain optimization, and quality control improvements.
- Government: Smart city initiatives, crime prediction, traffic management, and policy-making.

4. Challenges in Big Data Analytics

- Data Privacy and Security: Handling vast amounts of sensitive data brings the risk of breaches and requires compliance with regulations like GDPR.
- Data Integration: Integrating data from multiple sources can be challenging, especially when data is in different formats or stored in different systems.
- Scalability: As data grows, tools and infrastructure must be scalable to manage processing and storage.
- Talent Gap: There is a significant shortage of professionals with the skills to manage and analyze big data.

5. Future of Big Data Analytics

The future of big data is evolving rapidly. Advancements in artificial intelligence (AI) and machine learning (ML) are automating many aspects of big data analysis, making it easier to generate insights. Moreover, the integration of the Internet of Things (IoT) and edge computing is expected to expand the sources and velocity of data, further pushing the boundaries of big data analytics. Quantum computing is also on the horizon, potentially providing immense processing power for more complex data analysis.

What is Big Data Analytics ? – Definition, Working, Benefits

Big data analysis uses advanced analytical methods that can extract important business insights from bulk datasets. Within these datasets lies both structured (organized) and unstructured (unorganized) data. Its applications cover different industries such as healthcare, education, insurance, AI, retail, and manufacturing. By analyzing this data, organizations get better insight on what is good and what is bad, so they can make the necessary improvements, develop the production system, and increase profitability.

What is Big- Data Analytics?

This guide will discuss in greater detail the concept of big data analytics and how it impacts the decision making process in many parts of the corporate world. You will also know the different types of analyses that are used in big data, the list of the commonly used tools and the courses that can be recommended for you to start your journey towards the data analytics career

What is Big-Data Analytics?

Big data analytics is all about crunching massive amounts of information to uncover hidden trends, patterns, and relationships. It's like sifting through a giant mountain of data to find the gold nuggets of insight.

Here's a breakdown of what it involves:

- **Collecting Data:** Such data is coming from various sources such as social media, web traffic, sensors and customer reviews.
- **Cleaning the Data:** Imagine having to assess a pile of rocks that included some gold pieces in it. You would have to clean the dirt and the debris first. When data is being cleaned, mistakes must be fixed, duplicates must be removed and the data must be formatted properly.
- **Analyzing the Data:** It is here that the wizardry takes place. Data analysts employ powerful tools and techniques to discover patterns and trends. It is the same thing as looking for a specific pattern in all those rocks that you sorted through.

The multi-industrial utilization of big data analytics spans from healthcare to finance to retail. Through their data, companies can make better decisions, become more efficient, and get a competitive advantage.

How does big data analytics work?

Big Data Analytics is a powerful tool which helps to find the potential of large and complex datasets. To get better understanding, let's break it down into key steps:

- **Data Collection:** Data is the core of Big Data Analytics. It is the gathering of data from different sources such as the customers' comments, surveys, sensors, social media, and so on. The primary aim of data collection is to compile as much accurate data as possible. The more data, the more insights.
- **Data Cleaning (Data Preprocessing):** The next step is to process this information. It often requires some cleaning. This entails the replacement of missing data, the correction of inaccuracies, and the removal of duplicates. It is like sifting through a treasure trove, separating the rocks and debris and leaving only the valuable gems behind.
- **Data Processing:** After that we will be working on the data processing. This process contains such important stages as writing, structuring, and formatting of data in a way it will be usable for the analysis. It is like a chef who is gathering the ingredients before cooking. Data processing turns the data into a format suited for analytics tools to process.
- **Data Analysis:** Data analysis is being done by means of statistical, mathematical, and machine learning methods to get out the most important findings from the processed data. For example, it can uncover customer preferences, market trends, or patterns in healthcare data.
- **Data Visualization:** Data analysis usually is presented in visual form, for illustration – charts, graphs and interactive dashboards. The visualizations provided a way to simplify the large amounts of data and allowed for decision makers to quickly detect patterns and trends.
- **Data Storage and Management:** The stored and managed analyzed data is of utmost importance. It is like digital scrapbooking. May be you would want to go back to those lessons in the long run, therefore, how you store them has great importance. Moreover, data protection and adherence to regulations are the key issues to be addressed during this crucial stage.

- **Continuous Learning and Improvement:** Big data analytics is a continuous process of collecting, cleaning, and analyzing data to uncover hidden insights. It helps businesses make better decisions and gain a competitive edge.

Types of Big Data Analytics

Big Data Analytics comes in many different types, each serving a different purpose:

1. **Descriptive Analytics:** This type helps us understand past events. In social media, it shows performance metrics, like the number of likes on a post.
2. **Diagnostic Analytics:** In Diagnostic analytics delves deeper to uncover the reasons behind past events. In healthcare, it identifies the causes of high patient re-admissions.
3. **Predictive Analytics:** Predictive analytics forecasts future events based on past data. Weather forecasting, for example, predicts tomorrow's weather by analyzing historical patterns.
4. **Prescriptive Analytics:** However, this category not only predicts results but also offers recommendations for action to achieve the best results. In e-commerce, it may suggest the best price for a product to achieve the highest possible profit.
5. **Real-time Analytics:** The key function of real-time analytics is data processing in real time. It swiftly allows traders to make decisions based on real-time market events.
6. **Spatial Analytics:** Spatial analytics is about the location data. In urban management, it optimizes traffic flow from the data under the sensors and cameras to minimize the traffic jam.
7. **Text Analytics:** Text analytics delves into the unstructured data of text. In the hotel business, it can use the guest reviews to enhance services and guest satisfaction.

These types of analytics serve different purposes, making data understandable and actionable. Whether it's for business, healthcare, or everyday life, **Big Data Analytics** provides a range of tools to turn data into valuable insights, supporting better decision-making.

Big Data Analytics Technologies and Tools

Big Data Analytics relies on various technologies and tools that might sound complex, let's simplify them:

- **Hadoop:** Imagine Hadoop as an enormous digital warehouse. It's used by companies like Amazon to store tons of data efficiently. For instance, when Amazon suggests products you might like, it's because Hadoop helps manage your shopping history.
- **Spark:** Think of [Spark](#) as the super-fast data chef. Netflix uses it to quickly analyze what you watch and recommend your next binge-worthy show.
- **NoSQL Databases:** NoSQL databases, like [MongoDB](#), are like digital filing cabinets that Airbnb uses to store your booking details and user data. These databases are famous because of their quick and flexible, so the platform can provide you with the right information when you need it.
- **Tableau:** Tableau is like an artist that turns data into beautiful pictures. The World Bank uses it to create interactive charts and graphs that help people understand complex economic data.

- **Python and R:** [Python](#) and [R](#) are like magic tools for data scientists. They use these languages to solve tricky problems. For example, Kaggle uses them to predict things like house prices based on past data.
- **Machine Learning Frameworks (e.g., TensorFlow):** In [Machine learning](#) frameworks are the tools who make predictions. Airbnb uses [TensorFlow](#) to predict which properties are most likely to be booked in certain areas. It helps hosts make smart decisions about pricing and availability.

These tools and technologies are the building blocks of Big Data Analytics and helps organizations gather, process, understand, and visualize data, making it easier for them to make decisions based on information.

Benefits of Big Data Analytics

Big Data Analytics offers a host of real-world advantages, and let's understand with examples:

1. **Informed Decisions:** Imagine a store like Walmart. Big Data Analytics helps them make smart choices about what products to stock. This not only reduces waste but also keeps customers happy and profits high.
2. **Enhanced Customer Experiences:** Think about Amazon. Big Data Analytics is what makes those product suggestions so accurate. It's like having a personal shopper who knows your taste and helps you find what you want.
3. **Fraud Detection:** Credit card companies, like MasterCard, use Big Data Analytics to catch and stop fraudulent transactions. It's like having a guardian that watches over your money and keeps it safe.
4. **Optimized Logistics:** FedEx, for example, uses Big Data Analytics to deliver your packages faster and with less impact on the environment. It's like taking the fastest route to your destination while also being kind to the planet.

Challenges of Big data analytics

While Big Data Analytics offers incredible benefits, it also comes with its set of challenges:

- **Data Overload:** Consider Twitter, where approximately 6,000 tweets are posted every second. The challenge is sifting through this avalanche of data to find valuable insights.
- **Data Quality:** If the input data is inaccurate or incomplete, the insights generated by Big Data Analytics can be flawed. For example, incorrect sensor readings could lead to wrong conclusions in weather forecasting.
- **Privacy Concerns:** With the vast amount of personal data used, like in Facebook's ad targeting, there's a fine line between providing personalized experiences and infringing on privacy.
- **Security Risks:** With cyber threats increasing, safeguarding sensitive data becomes crucial. For instance, banks use Big Data Analytics to detect fraudulent activities, but they must also protect this information from breaches.

- **Costs:** Implementing and maintaining Big Data Analytics systems can be expensive. Airlines like Delta use analytics to optimize flight schedules, but they need to ensure that the benefits outweigh the costs.

Overcoming these challenges is essential to fully harness the power of Big Data Analytics. Businesses and organizations must tread carefully, ensuring they make the most of the insights while addressing these obstacles effectively.

Usage of Big Data Analytics

Big Data Analytics has a significant impact in various sectors:

- **Healthcare:** It aids in precise diagnoses and disease prediction, elevating patient care.
- **Retail:** Amazon's use of Big Data Analytics offers personalized product recommendations based on your shopping history, creating a more tailored and enjoyable shopping experience.
- **Finance:** Credit card companies such as Visa rely on Big Data Analytics to swiftly identify and prevent fraudulent transactions, ensuring the safety of your financial assets.
- **Transportation:** Companies like Uber use Big Data Analytics to optimize drivers' routes and predict demand, reducing wait times and improving overall transportation experiences.
- **Agriculture:** Farmers make informed decisions, boosting crop yields while conserving resources.
- **Manufacturing:** Companies like General Electric (GE) use Big Data Analytics to predict machinery maintenance needs, reducing downtime and enhancing operational efficiency.

Introduction to Data Science

In a world of data space where organizations deal with petabytes and exabytes of data, the era of Big Data emerged, and the essence of its storage also grew. It was a great challenge and concern for industries for the storage of data until 2010. Now when frameworks like Hadoop and others solved the problem of storage, the focus shifted to the processing of data. Data Science plays a big role here. All those fancy Sci-fi movies you love to watch around can be turned into reality by Data Science. Nowadays its growth has been increased in multiple ways and thus one should be ready for our future by learning what it is and how can we add value to it.

Introduction to data science

What is Data Science?

Data science is a multidisciplinary field that uses statistical and computational methods to extract insights and knowledge from data. It involves a combination of skills and knowledge from various

fields such as statistics, computer science, mathematics, and domain expertise. Data Science is kinda blended with various tools, algorithms, and machine learning principles. Most simply, it involves obtaining meaningful information or insights from structured or unstructured data through a process of analyzing, programming, and business skills. It is a field containing many elements like mathematics, statistics, computer science, etc. Those who are good at these respective fields with enough knowledge of the domain in which you are willing to work can call themselves as Data Scientist. It's not an easy thing to do but not impossible too. You need to start from data, it's visualization, programming, formulation, development, and deployment of your model. In the future, there will be great hype for data scientist jobs. Taking in that mind, be ready to prepare yourself to fit in this world.

Data science is a field that involves using statistical and computational techniques to extract insights and knowledge from data. It is a multi-disciplinary field that encompasses aspects of computer science, statistics, and domain-specific expertise. Data scientists use a variety of tools and methods, such as machine learning, statistical modeling, and data visualization, to analyze and make predictions from data. They work with both structured and unstructured data, and use the insights gained to inform decision making and support business operations. Data science is applied in a wide range of industries, including finance, healthcare, retail, and more. It helps organizations to make data-driven decisions and gain a competitive advantage.

How Data Science Works?

How data science works

Data science is not a one-step process such that you will get to learn it in a short time and call ourselves a Data Scientist. It's passes from many stages and every element is important. One should always follow the proper steps to reach the ladder. Every step has its value and it counts in your model. Buckle up in your seats and get ready to learn about those steps.

1. Problem Statement:

No work start without motivation, Data science is no exception though. It's really important to declare or formulate your problem statement very clearly and precisely. Your whole model and it's working depend on your statement. Many scientist considers this as the main and much important step of Date Science. So make sure what's your problem statement and how well can it add value to business or any other organization.

2. Data Collection:

After defining the problem statement, the next obvious step is to go in search of data that you might require for your model. You must do good research, find all that you need. Data can be in any form i.e unstructured or structured. It might be in various forms like videos, spreadsheets, coded forms, etc. You must collect all these kinds of sources.

3. Data Cleaning:

As you have formulated your motive and also you did collect your data, the next step to do is cleaning. Yes, it is! Data cleaning is the most favorite thing for data scientists to do. Data cleaning is all about the removal of missing, redundant, unnecessary and duplicate data from your collection. There are

various tools to do so with the help of programming in either R or [Python](#). It's totally on you to choose one of them. Various scientist have their opinion on which to choose. When it comes to the statistical part, R is preferred over Python, as it has the privilege of more than 12,000 packages. While python is used as it is fast, easily accessible and we can perform the same things as we can in R with the help of various packages.

4. Data Analysis and Exploration:

It's one of the prime things in data science to do and time to get inner Holmes out. It's about analyzing the structure of data, finding hidden patterns in them, studying behaviors, visualizing the effects of one variable over others and then concluding. We can explore the data with the help of various graphs formed with the help of libraries using any programming language. In R, GGplot is one of the most famous models while Matplotlib in Python.

5. Data Modelling:

Once you are done with your study that you have formed from data visualization, you must start building a hypothesis model such that it may yield you a good prediction in future. Here, you must choose a good algorithm that best fit to your model. There different kinds of algorithms from regression to classification, SVM(Support vector machines), Clustering, etc. Your model can be of a [Machine Learning](#) algorithm. You train your model with the train data and then test it with test data. There are various methods to do so. One of them is the K-fold method where you split your whole data into two parts, One is Train and the other is test data. On these bases, you train your model.

6. Optimization and Deployment:

You followed each and every step and hence build a model that you feel is the best fit. But how can you decide how well your model is performing? This where optimization comes. You test your data and find how well it is performing by checking its accuracy. In short, you check the efficiency of the data model and thus try to optimize it for better accurate prediction. Deployment deals with the launch of your model and let the people outside there to benefit from that. You can also obtain feedback from organizations and people to know their need and then to work more on your model.

Advice for new data science students

- **Curiosity** : If you are not curious , you would not know what to do with the data .
- **Judgmental** : It is because if you do not have preconceived notions about the things you wouldn't know where to begin with .
- **Argumentative** : It is because if you can argument and if you can plead a case , at least you can start somewhere and then you can learn from data and then can modify your assumptions.
- Start by gaining a solid understanding of the basics of programming, statistics, and linear algebra.
- Learn the tools of the trade such as Python, R, and SQL. Familiarize yourself with the most popular libraries and frameworks like numpy, pandas, and scikit-learn.

- Practice, practice, practice. Participate in online coding challenges and hackathons to improve your skills and gain experience.
- Learn the basics of machine learning and familiarize yourself with the most popular algorithms.
- Read research papers and stay up-to-date with the latest developments in the field.
- Learn how to communicate your findings effectively. Being able to present your work in a clear and compelling way is just as important as the technical skills you possess.
- Build a portfolio of projects that showcase your skills and experience.
- Network with other data scientists and professionals in the field. Attend meetups and conferences, and connect with people on LinkedIn.
- Be curious, and don't be afraid to ask questions.
- Finally, don't be discouraged if you encounter challenges or roadblocks along the way. Learning to become a data scientist is a journey, and it takes time, effort, and dedication to succeed.

Advantages of data science:

- Improved decision-making: Data science can help organizations make better decisions by providing insights and predictions based on data analysis.
- Cost-effective: With the right tools and techniques, data science can help organizations reduce costs by identifying areas of inefficiency and optimizing processes.
- Innovation: Data science can be used to identify new opportunities for innovation and to develop new products and services.
- Competitive advantage: Organizations that use data science effectively can gain a competitive advantage by making better decisions, improving efficiency, and identifying new opportunities.
- Personalization: Data science can help organizations personalize their products or services to better meet the needs of individual customers.

Disadvantages of data science:

- Data quality: The accuracy and quality of the data used in data science can have a significant impact on the results obtained.
- Privacy concerns: The collection and use of data can raise privacy concerns, particularly if the data is personal or sensitive.
- Complexity: Data science can be a complex and technical field that requires specialized skills and expertise.

- Bias: Data science algorithms can be biased if the data used to train them is biased, which can lead to inaccurate results.
- Interpretation: Interpreting data science results can be challenging, particularly for non-technical stakeholders who may not understand the underlying assumptions and methods used.

Conclusion

Data science have tremendous impact in machining learning, AI, big data, predictive analytics and decision-making. With a marrying of mathematical models, algorithms, and subject-matter knowledge, DSM allows for the effective analysis of large datasets. Data Management is not just about working with data! It is about understanding the data and utilizing the output in a meaningful way.

Indeed, it is highly likely that as the number of big data-related job openings will only continue to rise, so will the number of Data Scientist jobs. The capability of adapting to and commanding different Data Science tools and methodologies will become mandatory for any company and/or researcher intending to remain viable and effective in the marketplace as well as in proposing and implementing its decision-making processes.

The future of Data Science is brighter still ahead since innovations such as artificial intelligence and binning data analytics are evidently the future.

Thus, Data Science as an essential and constantly developing discipline shapes the great many of contemporary global trends. It is therefore easier to suggest that anyone interested in the actualization of potentials of data in his/her field of study will need to embrace the principles of the methodologies.

Big Data Characteristics:

6V's of Big Data

In recent years, Big Data was defined by the “3Vs” but now there is “6Vs” of Big Data which are also termed as the characteristics of Big Data as follows:

1. Volume:

- The name ‘Big Data’ itself is related to a size which is enormous.
- Volume is a huge amount of data.
- To determine the value of data, size of data plays a very crucial role. If the volume of data is very large, then it is actually considered as a ‘Big Data’. This means whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data.
- Hence while dealing with Big Data it is necessary to consider a characteristic ‘Volume’.
- *Example:* In the year 2016, the estimated global mobile traffic was 6.2 Exabytes (6.2 billion GB) per month. Also, by the year 2020 we will have almost 40000 Exabytes of data.

2. Velocity:

- Velocity refers to the high speed of accumulation of data.

- In Big Data velocity data flows in from sources like machines, networks, social media, mobile phones etc.
- There is a massive and continuous flow of data. This determines the potential of data that how fast the data is generated and processed to meet the demands.
- Sampling data can help in dealing with the issue like 'velocity'.
- *Example:* There are more than 3.5 billion searches per day are made on Google. Also, Facebook users are increasing by 22%(Approx.) year by year.

3. Variety:

- It refers to nature of data that is structured, semi-structured and unstructured data.
- It also refers to heterogeneous sources.
- Variety is basically the arrival of data from new sources that are both inside and outside of an enterprise. It can be structured, semi-structured and unstructured.
 - **Structured data:** This data is basically an organized data. It generally refers to data that has defined the length and format of data.
 - **Semi- Structured data:** This data is basically a semi-organised data. It is generally a form of data that do not conform to the formal structure of data. Log files are the examples of this type of data.
 - **Unstructured data:** This data basically refers to unorganized data. It generally refers to data that doesn't fit neatly into the traditional row and column structure of the relational database. Texts, pictures, videos etc. are the examples of unstructured data which can't be stored in the form of rows and columns.

4. Veracity:

- It refers to inconsistencies and uncertainty in data, that is data which is available can sometimes get messy and quality and accuracy are difficult to control.
- Big Data is also variable because of the multitude of data dimensions resulting from multiple disparate data types and sources.
- *Example:* Data in bulk could create confusion whereas less amount of data could convey half or Incomplete Information.

5. Value:

- After having the 4 V's into account there comes one more V which stands for Value! The bulk of Data having no Value is of no good to the company, unless you turn it into something useful.

- Data in itself is of no use or importance but it needs to be converted into something valuable to extract Information. Hence, you can state that Value! is the most important V of all the 6V's.

6. Variability:

- How fast or available data that extent is the structure of your data is changing?
- How often does the meaning or shape of your data change?
- *Example:* if you are eating same ice-cream daily and the taste just keep changing.

Architecture of Big Data Systems

What is Big Data?

Data science is the study of data analysis by advanced technology ([Machine Learning](#), [Artificial Intelligence](#), Big data). It processes a huge amount of structured, semi-structured, and unstructured data to extract insight meaning, from which one pattern can be designed that will be useful to take a decision for grabbing the new business opportunity, the betterment of product/service, and ultimately business growth. Data science process to make sense of Big data/huge amount of data that is used in business. The workflow of Data science is as below:

- Objective and the issue of business determining – What is the organization's objective, what level the organization wants to achieve, and what issue the company is facing -these are the factors under consideration. Based on such factors which type of data are relevant is considered.
- Collection of relevant data- relevant data are collected from various sources.
- Cleaning and filtering collected data – non-relevant data are removed.
- Explore the filtered, cleaned data – Finding any hidden pattern, synchronization in data, plotting them in the graph, chart, etc. form that is understandable to a non-technical person.
- Creating a model by analyzing data – creating a model, validate it.
- Visualization of finding by interpreting data or creating a model for a business person.
- Help businesspeople in making the decision and taking the step for the sack of business growth.

Data Mining: It is a process of extracting insight meaning, hidden patterns from collected data that is useful to take a business decision for the purpose of decreasing expenditure and increasing revenue.

Big Data: This is a term related to extracting meaningful data by analyzing the huge amount of complex, variously formatted data generated at high speed, that cannot be handled, or processed by the traditional system. **Data Expansion Day by Day:** Day by day amount of data increasing exponentially because of today's various data production sources like a smart electronic devices. As per IDC (International Data Corporation) report, new data created per each person in the world per

second by 2020 will be 1.7 MB. The amount of total data in the world by 2020 will reach around 44 ZettaBytes (44 trillion GigaByte) and 175 ZettaBytes by 2025. It is being seen that the total volume of data is double every two years. The total size growth of data worldwide, year to year as per the IDC report is shown below:

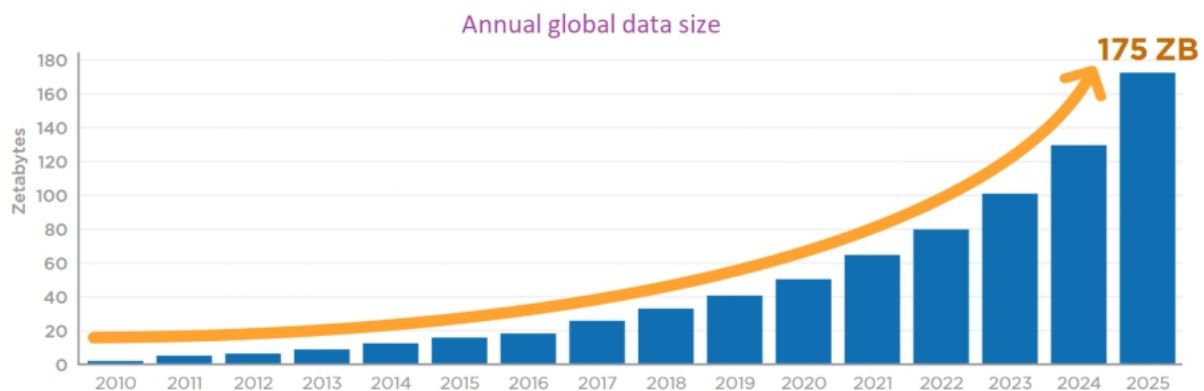


Image Source: Google

Source of Big Data:

- **Social Media:** Today's world a good percent of the total world population is engaged with social media like Facebook, WhatsApp, Twitter, YouTube, Instagram, etc. Each activity on such media like uploading a photo, or video, sending a message, making comment, putting like, etc create data.
- **A sensor placed in various places:** Sensor placed in various places of the city that gathers data on temperature, humidity, etc. A camera placed beside the road gather information about traffic condition and creates data. Security cameras placed in sensitive areas like airports, railway stations, and shopping malls create a lot of data.
- **Customer Satisfaction Feedback:** Customer feedback on the product or service of the various company on their website creates data. For Example, retail commercial sites like Amazon, Walmart, Flipkart, and Myntra gather customer feedback on the quality of their product and delivery time. Telecom companies, and other service provider organizations seek customer experience with their service. These create a lot of data.
- **IoT Appliance:** Electronic devices that are connected to the internet create data for their smart functionality, examples are a smart TV, smart washing machine, smart coffee machine, smart AC, etc. It is machine-generated data that are created by sensors kept in various devices. For Example, a Smart printing machine – is connected to the internet. A number of such printing machines connected to a network can transfer data within each other. So, if anyone loads a file copy in one printing machine, the system stores that file content, and another printing machine kept in another building or another floor can print out that file hard copy. Such data transfer between various printing machines generates data.
- **E-commerce:** In e-commerce transactions, business transactions, banking, and the stock market, lots of records stored are considered one of the sources of big data. Payments through credit cards, debit cards, or other electronic ways, all are kept recorded as data.

- **Global Positioning System (GPS):** GPS in the vehicle helps in monitoring the movement of the vehicle to shorten the path to a destination to cut fuel, and time consumption. This system creates huge data on vehicle position and movement.
- **Transactional Data:** Transactional data, as the name implies, is information obtained through online and offline transactions at various points of sale. The data contains important information about transactions, such as the date and time of the transaction, the location where it took place, the items bought, their prices, the methods of payment, the discounts or coupons that were applied, and other pertinent quantitative data. These are some of the sources of transactional data: orders for payment, Invoices, E-receipts and recordkeeping etc.
- **Machine Data:** Automatically generated machine data is produced in reaction to an event or according to a set timetable. This indicates that all of the data was compiled from a variety of sources, including satellites, desktop computers, mobile phones, industrial machines, smart sensors, SIEM logs, medical and wearable devices, road cameras, IoT devices, and more. Businesses can monitor consumer behaviour thanks to these sources. Data derived from automated sources expands exponentially in tandem with the market's shifting external environment. These sensors are used to capture this kind of information: In a broader sense, machine data includes data that is generated by servers, user applications, websites, cloud programmes, and other sources.

Data Architecture Design and Data Management

In the beginning times of computers and Internet, the data used was not as much of as it is today, The data then could be so easily stored and managed by all the users and business enterprises on a single computer, because the data never exceeded to the extent of 19 exabytes but now in this era, the data has increased about 2.5 quintillions per day.

Most of the data is generated from social media sites like Facebook, Instagram, Twitter, etc, and the other sources can be e-business, e-commerce transactions, hospital, school, bank data, etc. This data is impossible to manage by traditional data storing techniques. So Big-Data came into existence for handling the data which is big and impure.

[Big Data](#) is the field of collecting the large data sets from various sources like social media, GPS, sensors etc and analyzing them systematically and extract useful patterns using some [tools](#) and techniques by enterprises. Before analyzing and determining the data, the data architecture must be designed by the architect.

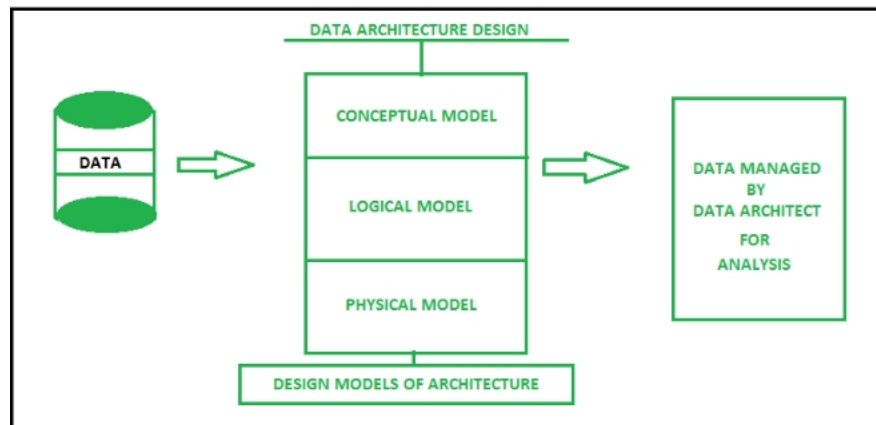
Data architecture Design and Data Management :
Data architecture design is set of standards which are composed of certain policies, rules, models and standards which manages, what type of data is collected, from where it is collected, the arrangement of collected data, storing that data, utilizing and securing the data into the systems and data warehouses for further analysis.

Data is one of the essential pillars of enterprise architecture through which it succeeds in the execution of business strategy.

Data architecture design is important for creating a vision of interactions occurring between data systems, like for example if data architect wants to implement data integration, so it will need

interaction between two systems and by using data architecture the visionary model of data interaction during the process can be achieved.

Data architecture also describes the type of data structures applied to manage data and it provides an easy way for data preprocessing. The data architecture is formed by dividing into three essential models and then are combined :



- **Conceptual model** –
It is a business model which uses [Entity Relationship \(ER\) model](#) for relation between entities and their attributes.
- **Logical model** –
It is a model where problems are represented in the form of logic such as rows and column of data, classes, xml tags and other DBMS techniques.
- **Physical model** –
Physical models holds the database design like which type of database technology will be suitable for architecture.

A data architect is responsible for all the design, creation, manage, deployment of data architecture and defines how data is to be stored and retrieved, other decisions are made by internal bodies.

Factors that influence Data Architecture :
Few influences that can have an effect on data architecture are business policies, business requirements, Technology used, economics, and data processing needs.

- **Business-requirements–**
These include factors such as the expansion of business, the performance of the system access, data management, transaction management, making use of raw data by converting them into image files and records, and then storing in data warehouses. Data warehouses are the main aspects of storing transactions in business.
- **Business-policies–**
The policies are rules that are useful for describing the way of processing data. These policies are made by internal organizational bodies and other government agencies.
- **Technology in use–**
This includes using the example of previously completed data architecture design and also using existing licensed software purchases, database technology.

- **Business-economics–**
The economical factors such as business growth and loss, interest rates, loans, condition of the market, and the overall cost will also have an effect on design architecture.
- **Data processing needs–**
These include factors such as mining of the data, large continuous transactions, database management, and other data preprocessing needs.

Data Management :

- Data management is the process of managing tasks like extracting data, storing data, transferring data, processing data, and then securing data with low-cost consumption.
- Main motive of data management is to manage and safeguard the people's and organization data in an optimal way so that they can easily create, access, delete, and update the data.
- Because data management is an essential process in each and every enterprise growth, without which the policies and decisions can't be made for business advancement. The better the data management the better productivity in business.
- Large volumes of data like big data are harder to manage traditionally so there must be the utilization of optimal technologies and tools for data management such as Hadoop, Scala, Tableau, AWS, etc. Which can further used for big data analysis in achieving improvements in patterns.
- Data management can be achieved by training the employees necessarily and maintenance by DBA, data analyst, and data architects.

Advantages of Big Data:

Big Data is “data of a very large size, typically to the extent that its manipulation and management present logistical challenges”. Big Data refers to technologies and initiatives that involve data that is too diverse, fast-changing, or massive for conventional technologies, skills, and infrastructure to address efficiently. But nowadays with the help of new technologies, it is very easy to realize the value of Big data, for example, to identify shopping behavioral trends of customers to improve stockage and pricing of the items. Government processes also get benefits and banking institutions are capturing data on customer interaction to model risk and fraud.

What is Big Data?

Big Data is a term that is concerned with the collection of huge data associated with characteristics like large variety, more velocity along with increased volume simply known as 3 Vs. There are mainly three types of Big Data – Structured data, Unstructured data, and Semi-structured data. This voluminous structured, semi-structured, or unstructured data is processed to gather useful insights for subsequently making effective business decisions, etc.

Use of Big Data:

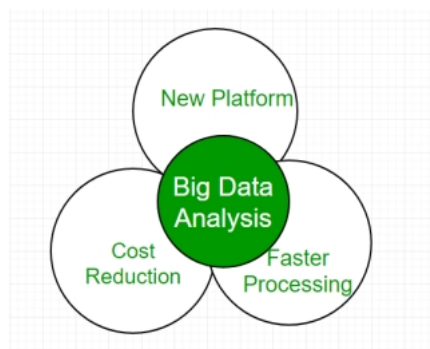
Big Data is used for many things but some of the main usage purposes are as follows:

- Big Data enables you to gather information about customers and their experience, then eventually helps you to align it properly.

- Helps in maintaining the predictive failures beforehand by analyzing the problems and provides with their potential solutions.
- Big Data is also useful for companies to anticipate customer demand, roll out new plans, test markets, etc.
- Big Data is very useful in assessing predictive failures by analyzing various indicators such as unstructured data, error messages, log entries, engine temperature, etc.
- Big Data is also very efficient in maintaining operational functions along with anticipating future demands of the customers, current market demands thus providing proper results.

Benefits of Big Data in IT Sectors:

- Many old IT companies are fully dependent on big data in order to modernize their outdated mainframes by identifying the root causes of failures and issues in real-time and antiquated code bases. Many organizations are replacing their traditional system with open-source platforms like Hadoop.
- Most big data solutions are based on [Hadoop](#), which allows designs to scale up from a single machine to thousands of machines, each offering local computation and storage, moreover, it is a “free” open source platform, allowing minimizing capital investment for an organization in acquiring new platforms.
- With the help of big data technologies IT companies are able to process third-party data fast, which is often hard to understand at once by having inherently high horsepower and parallelized working of platforms.



Benefits of Big Data in Business:

- Data quality has a direct impact on business process efficiency. In purchase to pay process, poor quality vendor data can cause missing purchase contracts or pricing information which can lead to delays in procuring vital goods. Many companies use big data solutions or algorithms to simply do what they have already been doing, so that there is no data loss moreover if we run the algorithm against the data set, the result might be the list of individual who exhibits attributes of fraudulent behavior.
- In order to cash process, incomplete or inaccurate credit limits or pricing information can lead to overall customer service loss or reduce revenue or may increase service cost, with the help of big data technologies and the ability to run various algorithms more quickly, the data can be updated at regular intervals throughout the day.

- The systematic analysis of data or data profiling is used to assess the overall health of the data which leads to proper business decisions in accordance with the present situation because sometimes inaccurate data results in incorrect management, which means business decisions are based on incorrect information. For example, the more one can understand customers' complex relationships, preferences, and interaction history with the company, the more relevant and timely business outreach.



Benefits of Big Data in Enterprise :

- Big data might allow a company to collect trillions or billions of real-time data points on its products, resources, or customers- and then repackage the data instantaneously to optimize the customer experience.
- The speed at which data is updated using big data technologies allows enterprises to more quickly and accurately respond to customer demands. For example, MetLife used MongoDB to quickly consolidate customer information over 70 different sources and provide a single, rapidly-updated view
- Big data can help enterprises to act more nimbly allowing them to adapt to changes faster than their competitors.



Benefits of Big Data in Other Areas:

- Big Data technologies are used to predict the 'buy' and 'sell' decisions made on the shares of different companies for the customers.

- Hospitals are analyzing medical data and records to predict those patients that are likely to seek re-admission within a few months of discharge. The hospital can avoid costly stays for patients in the hospital.
- Search-Engine retrieves lots of Data from different databases in a fraction of a second using big data technologies. For example, Google uses the MapReduce algorithm to search for a given query. MapReduce divides the task into small parts and assigns those parts to many computers connected over the network, and collects the result to form the final result.
- Financial Services organizations are using big data for data mining about customer interactions to slice and dice their users into finely tuned segments, this will help in creating increasingly relevant and sophisticated offers.
- Insurance companies are using Big Data analysis to see which home insurance applications can be immediately processed, and which ones need a validating in-person visit from an agent.
- Web-based businesses are developing information products that combine data gathered from customers to offer more appealing recommendations and more successful coupon programs.

Challenges faced by Big Data Systems

The *Challenges in Big Data* are the real implementation hurdles. These require immediate attention and need to be handled because if not handled, the technology's failure may occur, leading to some unpleasant results. Big data challenges include storing and analyzing extremely large and fast-growing data.

Big Challenges with Big Data

This article explores *some of the most pressing challenges associated with Big Data and offers potential solutions for overcoming them.*

Big Challenges with Big Data

- [Data Volume: Managing and Storing Massive Amounts of Data](#)
- [Data Variety: Handling Diverse Data Types](#)
- [Data Velocity: Processing Data in Real-Time](#)
- [Data Veracity: Ensuring Data Quality and Accuracy](#)
- [Data Security and Privacy: Protecting Sensitive Information](#)
- [Data Integration: Combining Data from Multiple Sources](#)
- [Data Analytics: Extracting Valuable Insights](#)
- [Data Governance: Establishing Policies and Standards](#)

Data Volume: Managing and Storing Massive Amounts of Data

- **Challenge:** The most apparent challenge with [Big Data](#) is the sheer volume of data being generated. Organizations are now dealing with petabytes or even exabytes of data, making

traditional storage solutions inadequate. This vast amount of data requires advanced storage infrastructure, which can be costly and complex to maintain.

- **Solution:** Adopting scalable cloud storage solutions, such as *Amazon S3*, *Google Cloud Storage*, or *Microsoft Azure*, can help manage large volumes of data. These platforms offer flexible storage options that can grow with your data needs. Additionally, implementing data compression and deduplication techniques can reduce storage costs and optimize the use of available storage space.

Data Variety: Handling Diverse Data Types

- **Challenge:** Big Data encompasses a wide variety of data types, including structured data (e.g., databases), semi-structured data (e.g., XML, JSON), and unstructured data (e.g., text, images, videos). The diversity of [data types](#) can make it difficult to integrate, analyze, and extract meaningful insights.
- **Solution:** To address the challenge of data variety, organizations can employ data integration platforms and tools like Apache Nifi, Talend, or Informatica. These tools help in consolidating disparate data sources into a unified data model. Moreover, adopting schema-on-read approaches, as opposed to traditional schema-on-write, allows for more flexibility in handling diverse data types.

Data Velocity: Processing Data in Real-Time

- **Challenge:** The speed at which data is generated and needs to be processed is another significant challenge. For instance, [IoT](#) devices, social media platforms, and financial markets produce data streams that require real-time or near-real-time processing. Delays in processing can lead to missed opportunities and inefficiencies.
- **Solution:** To handle high-velocity data, organizations can implement real-time data processing frameworks such as [Apache Kafka](#), [Apache Flink](#), or Apache Storm. These frameworks are designed to handle high-throughput, low-latency data processing, enabling businesses to react to events as they happen. Additionally, leveraging edge computing can help process data closer to its source, reducing latency and improving real-time decision-making.

Data Veracity: Ensuring Data Quality and Accuracy

- **Challenge:** With Big Data, ensuring the quality, accuracy, and reliability of data—referred to as data veracity—becomes increasingly difficult. Inaccurate or low-quality data can lead to misleading insights and poor decision-making. Data veracity issues can arise from various sources, including data entry errors, inconsistencies, and incomplete data.
- **Solution:** Implementing robust data governance frameworks is crucial for maintaining data veracity. This includes establishing data quality standards, performing regular data audits, and employing data cleansing techniques. Tools like Trifacta, Talend Data Quality, and Apache Griffin can help automate and streamline data quality management processes.

Data Security and Privacy: Protecting Sensitive Information

- **Challenge:** As organizations collect and store more data, they face increasing risks related to data security and privacy. High-profile data breaches and growing concerns over data privacy regulations, such as GDPR and CCPA, highlight the importance of safeguarding sensitive information.
- **Solution:** To mitigate security and privacy risks, organizations must adopt comprehensive data protection strategies. This includes implementing encryption, access controls, and regular security audits. Additionally, organizations should stay informed about evolving data privacy regulations and ensure compliance by adopting privacy-by-design principles in their data management processes.

Data Integration: Combining Data from Multiple Sources

- **Challenge:** Integrating data from various sources, especially when dealing with legacy systems, can be a daunting task. Data silos, where data is stored in separate systems without easy access, further complicate the integration process, leading to inefficiencies and incomplete analysis.
- **Solution:** [Data integration](#) platforms like [Apache Camel](#), MuleSoft, and IBM DataStage can help streamline the process of integrating data from multiple sources. Adopting a microservices architecture can also facilitate easier data integration by breaking down monolithic applications into smaller, more manageable services that can be integrated more easily.

Data Analytics: Extracting Valuable Insights

- **Challenge:** The ultimate goal of Big Data is to derive actionable insights, but the complexity of analyzing large, diverse datasets can be overwhelming. Traditional analytical tools may struggle to scale, and the lack of skilled data scientists can further hinder the ability to extract meaningful insights.
- **Solution:** Organizations should invest in advanced analytics platforms like [Apache Spark](#), [Hadoop](#), or [Google BigQuery](#), which are designed to handle large-scale data processing and analysis. Additionally, fostering a culture of data literacy and providing training for employees can help bridge the skills gap and empower teams to effectively analyze Big Data.

Data Governance: Establishing Policies and Standards

- **Challenge:** As data becomes a critical asset, establishing effective data governance becomes essential. However, many organizations struggle with creating and enforcing policies and standards for data management, leading to issues with data consistency, quality, and compliance.
- **Solution:** Implementing a formal data governance framework is key to overcoming this challenge. This framework should define roles and responsibilities, establish data stewardship programs, and enforce data management policies.